DHASARATHAN THANGARAJ

Full Stack Developer

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Summary

Full Stack Developer with 3 years of experience building scalable web applications using Node.js, Express.js, React.js, Python, and Django. Proficient in backend development, RESTful API design, frontend development (React.js, Angular), JWT/RBAC security, and database design & optimization (MySQL, MS SQL Server, PostgreSQL + pgvector). Experienced in LLM integration and Generative AI through a production-grade Advanced RAG pipeline. Skilled in CI/CD (GitHub Actions), Docker, Kubernetes, microservices, Linux, system design, and Git within Agile/Scrum environments.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL
Frontend: React.js, Redux, Angular, RxJS
Backend: Node.js, Express.js, Django, Django REST Framework, REST APIs, Microservices
Databases: MySQL, MS SQL Server, PostgreSQL (pgvector)
DevOps / CI-CD: Docker, Linux, Kubernetes, Helm, GitHub Actions, Git, SonarQube
AI / Tools: OpenAI API, Gemini, LangChain, JWT / RBAC, Postman, MkDocs

Work Experience

Infinite Services July 2026 – Nov 2026
Software Consultant

- Built full-stack web applications using Node.js, Express.js, React.js, TypeScript, and Python/Django with RESTful API design, within an Agile/Scrum team.
- Managed CI/CD pipelines with GitHub Actions, Docker containerisation, and Kubernetes (Helm) microservices deployments.
- Enforced code quality with SonarQube, maintained Git branching workflows and MkDocs documentation, and contributed to system and API design discussions.

Independent Freelancer Dec 2025 – June 2026
Full Stack Developer

- Developed full-stack web applications using React.js, Node.js, Express.js, and Python/Django, integrating REST APIs and SQL databases.
- Containerised applications with Docker and automated delivery using GitHub Actions, maintaining code quality, Git workflows, and documentation.

Agile Civil Engg. July 2023 – Nov 2025
Software Engineer

- Developed and optimised Django REST APIs and MS SQL Server databases, improving response time by 30% and data retrieval efficiency by 40%.
- Built dynamic Angular frontends with RxJS state management; implemented JWT authentication and RBAC for enhanced application security.
- Conducted API testing with Postman and followed Agile/Scrum methodologies for iterative, on-time delivery.

Sparks Foundation, Singapore Jan 2023
Software Engineer Intern

- Analysed and cleaned large datasets using Python (Pandas, NumPy), improving data quality by 25% and cutting manual processing effort by 20% through scripted automation.
- Built data visualisations and reports in Python to surface business insights for stakeholders.

Projects

AI Document Assistant – Advanced RAG Application github.com/dhas97
React 18 + Vite, Django 4.2 + DRF, Gemini 2.5 Flash, PostgreSQL + pgvector, Docker Compose

- Engineered an 8-stage RAG pipeline: query rewriting, multi-query generation, HyDE, hybrid retrieval (BM25 + parallel pgvector HNSW ANN searches), Reciprocal Rank Fusion, Gemini-as-judge reranking, and grounded generation with inline citations.
- Implemented parent-child semantic chunking (NLTK) and task-typed Gemini embeddings (768-dim) stored in PostgreSQL 16 with a pgvector HNSW index for fast ANN queries.
- Delivered structured citations (inline markers, source cards with page numbers) and opt-in RAG evaluation metrics: Faithfulness, Answer Relevancy, and Context Relevancy.

AI Data Agent – Natural-Language-to-SQL Assistant
LangChain, LLM, Text-to-SQL, Python

- Built an AI data agent that turns plain-English questions into database queries, executes them, and returns readable answers, removing the need to write SQL manually.

Certifications & Courses

Data Analytics – Coursera | **Project Management** – Coursera | **SQL (Advanced)** – HackerRank

Education

B.E. Civil Engineering, KLN College of Information Technology, Madurai 2015 – 2019